

Electric charge and Electric current

1. Electric charge is of two types: _____ and _____.

- a) positive, negative
- b) proton, electron
- c) north, south
- d) AC, DC

Correct answer: a) positive, negative

2. Like charges _____.

- a) attract
- b) repel
- c) neutralise
- d) discharge

Correct answer: b) repel

3. SI unit of electric charge is _____.

- a) ampere
- b) volt
- c) coulomb
- d) ohm

Correct answer: c) coulomb

4. 1 coulomb = _____ electrons.

- a) 6.25×10^{18}
- b) 6.023×10^{23}
- c) 10^{12}
- d) 10^6

Correct answer: a) 6.25×10^{18}

5. Electric potential is measured in _____.

- a) joule
- b) ampere
- c) volt
- d) watt

Correct answer: c) volt

6. Potential difference $V =$ _____.

- a) W/q
- b) q/W
- c) $I \times R$
- d) P/I

Correct answer: a) W/q

7. Electric current $I =$ _____.

- a) Q/t
- b) V/R
- c) P/V
- d) all

Correct answer: a) Q/t

8. SI unit of current is _____.

- a) volt
- b) ampere
- c) coulomb
- d) ohm

Correct answer: b) ampere

9. Conventional current flows from _____.

- a) positive to negative

b) negative to positive

c) electron to proton

d) no direction

Correct answer: a) positive to negative

10. Electron current flows from _____.

a) positive to negative

b) negative to positive

c) same as conventional

d) random

Correct answer: b) negative to positive

11. Electromotive force (emf) is _____.

a) work per unit charge

b) current

c) resistance

d) power

Correct answer: a) work per unit charge

12. Resistance R depends on _____.

a) length

b) area

c) material

d) all

Correct answer: d) all

13. Ohm's law: $V =$ _____.

a) IR

b) I/R

c) R/I

d) I^2R

Correct answer: a) IR

14. SI unit of resistance is _____.

a) ampere

b) volt

c) ohm

d) watt

Correct answer: c) ohm

15. Resistivity $\rho =$ _____.

a) RA/I

b) I/RA

c) RI/A

d) AI/R

Correct answer: a) RA/I

16. Conductors have low _____.

a) resistance

b) resistivity

c) emf

d) both a and b

Correct answer: d) both a and b

17. Insulators have high _____.

a) resistance

b) current

c) emf

d) power

Correct answer: a) resistance

18. Superconductors have _____.

- a) zero resistance
- b) infinite resistance
- c) high resistance
- d) variable

Correct answer: a) zero resistance

19. Factors affecting resistance: directly proportional to _____.

- a) length
- b) temperature (metals)
- c) resistivity
- d) all

Correct answer: d) all

20. Electric power $P =$ _____.

- a) VI
- b) V/I
- c) I/V
- d) IR

Correct answer: a) VI

21. Unit of power is _____.

- a) joule
- b) watt
- c) ampere
- d) volt

Correct answer: b) watt

22. $P = I^2R$ or _____.

- a) V^2/R

b) both

c) V^2R

d) I/V

Correct answer: b) both

23. Heating effect used in _____.

a) fuse

b) heater

c) bulb

d) all

Correct answer: d) all

24. Chemical effect: _____.

a) electrolysis

b) electroplating

c) both

d) heating

Correct answer: c) both

25. Magnetic effect: _____.

a) electromagnet

b) motor

c) both

d) fuse

Correct answer: c) both

26. Electric circuit must be _____.

a) closed

b) open

c) short

d) parallel

Correct answer: a) closed

27. Ammeter connected in _____.

a) series

b) parallel

c) either

d) none

Correct answer: a) series

28. Voltmeter connected in _____.

a) series

b) parallel

c) both

d) none

Correct answer: b) parallel

29. Domestic circuits: live wire is _____.

a) red/brown

b) black

c) green

d) yellow

Correct answer: a) red/brown

30. Neutral wire is _____.

a) blue

b) green

c) red

d) yellow-green

Correct answer: a) blue

31. Earth wire is _____.

- a) green
- b) yellow
- c) green-yellow
- d) red

Correct answer: c) green-yellow

32. Fuse wire has _____.

- a) high melting point
- b) low melting point
- c) high resistance
- d) b and c

Correct answer: d) b and c

33. MCB stands for _____.

- a) Miniature Circuit Breaker
- b) Main Circuit Board
- c) Magnetic Circuit Breaker
- d) Manual Circuit Breaker

Correct answer: a) Miniature Circuit Breaker

34. AC frequency in India _____.

- a) 50 Hz
- b) 60 Hz
- c) 100 Hz
- d) 25 Hz

Correct answer: a) 50 Hz

35. DC flows in _____.

- a) one direction

b) both directions

c) varying

d) zero

Correct answer: a) one direction

36. Bird on high voltage line safe because _____.

a) no potential difference

b) high resistance

c) feathers

d) small size

Correct answer: a) no potential difference

37. Earthing prevents _____.

a) shock

b) current

c) voltage

d) power loss

Correct answer: a) shock

38. 1 kWh = _____.

a) 3.6×10^6 J

b) 10^6 J

c) 3600 J

d) 1000 J

Correct answer: a) 3.6×10^6 J

39. Commercial unit of energy _____.

a) kWh

b) joule

c) watt

d) volt

Correct answer: a) kWh

40. Series connection: current same, voltage _____.

a) divides

b) same

c) multiplies

d) zero

Correct answer: a) divides

41. Parallel connection: voltage same, current _____.

a) divides

b) same

c) adds

d) zero

Correct answer: c) adds

42. Reciprocal of resistance is _____.

a) conductance

b) resistivity

c) current

d) power

Correct answer: a) conductance

43. Alloys have higher _____.

a) resistance

b) conductivity

c) emf

d) power

Correct answer: a) resistance

44. Temperature coefficient for metals is _____.

- a) positive
- b) negative
- c) zero
- d) infinite

Correct answer: a) positive

45. Semiconductors have _____.

- a) negative coefficient
- b) positive
- c) zero
- d) variable

Correct answer: a) negative

46. Kirchhoff's current law: sum of currents _____.

- a) zero
- b) maximum
- c) minimum
- d) equal

Correct answer: a) zero

47. Kirchhoff's voltage law: sum of voltages _____.

- a) zero
- b) positive
- c) negative
- d) infinite

Correct answer: a) zero

48. LED works on _____.

- a) forward bias

b) reverse bias

c) no bias

d) AC

Correct answer: a) forward bias

49. Diode allows current in _____.

a) one direction

b) both

c) neither

d) varying

Correct answer: a) one direction

50. Transistor has _____.

a) two junctions

b) one junction

c) three junctions

d) none

Correct answer: a) two junctions

51. NPN transistor: emitter is _____.

a) negative

b) positive

c) neutral

d) variable

Correct answer: a) negative

52. Logic gates: AND gate output 1 if _____.

a) all 1

b) any 1

c) all 0

d) any 0

Correct answer: a) all 1

53. OR gate output 1 if _____.

a) all 1

b) any 1

c) all 0

d) none

Correct answer: b) any 1

54. NOT gate inverts _____.

a) input

b) output

c) both

d) none

Correct answer: a) input

55. Safety device in homes _____.

a) fuse

b) MCB

c) earthing

d) all

Correct answer: d) all

56. Overload causes _____.

a) heating

b) short circuit

c) both

d) none

Correct answer: c) both

57.Short circuit: resistance _____.

- a) zero
- b) high
- c) infinite
- d) normal

Correct answer: a) zero

58.Human body resistance dry _____.

- a) 100,000 Ω
- b) 1000 Ω
- c) 1 M Ω
- d) 10 Ω

Correct answer: a) 100,000 Ω

59.Wet skin resistance _____.

- a) lower
- b) higher
- c) same
- d) zero

Correct answer: a) lower

60.To avoid shock, hold plug by _____.

- a) pins
- b) body
- c) wires
- d) none

Correct answer: b) body

61.Voltage rating of house supply in India _____.

- a) 220 V

b) 110 V

c) 240 V

d) 100 V

Correct answer: a) 220 V

62. Frequency 50 Hz means _____.

a) 50 cycles/sec

b) 50 V

c) 50 A

d) 50 W

Correct answer: a) 50 cycles/sec

63. Advantages of AC: easily _____.

a) transformed

b) stored

c) both

d) none

Correct answer: c) both

64. DC used in _____.

a) batteries

b) electronics

c) both

d) homes

Correct answer: c) both

65. RMS value for AC _____.

a) 0.707 peak

b) peak

c) average

d) zero

Correct answer: a) 0.707 peak

66. Electric bulb filament _____.

a) tungsten

b) copper

c) iron

d) aluminium

Correct answer: a) tungsten

67. Fuse material _____.

a) tin-lead alloy

b) copper

c) silver

d) gold

Correct answer: a) tin-lead alloy

68. 5 A fuse for _____.

a) lights

b) heaters

c) AC

d) fans

Correct answer: a) lights

69. Series: equivalent $R =$ _____.

a) sum

b) reciprocal sum

c) product

d) divide

Correct answer: a) sum

70.Parallel: $1/R_{eq} =$ _____.

- a) sum $1/R$
- b) sum R
- c) product
- d) average

Correct answer: a) sum $1/R$

71.Wheatstone bridge used for _____.

- a) unknown R
- b) current
- c) voltage
- d) power

Correct answer: a) unknown R

72.Potentiometer measures _____.

- a) emf accurately
- b) current
- c) resistance
- d) power

Correct answer: a) emf accurately

73.Galvanometer detects _____.

- a) small current
- b) voltage
- c) power
- d) resistance

Correct answer: a) small current

74.Shunt used to _____.

- a) increase range ammeter

- b) decrease
- c) measure voltage
- d) power

Correct answer: a) increase range ammeter

75. Multiplier for _____.

- a) voltmeter
- b) ammeter
- c) galvanometer
- d) all

Correct answer: a) voltmeter

76. Cell internal resistance _____.

- a) reduces terminal voltage
- b) increases
- c) no effect
- d) zero

Correct answer: a) reduces terminal voltage

77. Cells in series: voltage _____.

- a) adds
- b) divides
- c) same
- d) zero

Correct answer: a) adds

78. Cells in parallel: current _____.

- a) adds
- b) divides
- c) voltage adds

d) resistance adds

Correct answer: a) adds

79. Charging by friction: glass rod + silk becomes _____.

a) positive

b) negative

c) neutral

d) grounded

Correct answer: a) positive

80. Ebonite rod + fur becomes _____.

a) positive

b) negative

c) neutral

d) positive

Correct answer: b) negative

81. Electroscope detects _____.

a) charge

b) current

c) voltage

d) resistance

Correct answer: a) charge

82. Gold leaf electroscope has _____.

a) gold leaves

b) copper

c) aluminium

d) iron

Correct answer: a) gold leaves

83.Charging by conduction: charged body touches _____.

- a) electroscope
- b) discharges
- c) both
- d) none

Correct answer: c) both

84.Charging by induction: bring _____.

- a) opposite charge near
- b) touch ground
- c) both
- d) friction

Correct answer: c) both

85.Capacitor stores _____.

- a) charge
- b) current
- c) voltage
- d) power

Correct answer: a) charge

86.Dielectric increases _____.

- a) capacitance
- b) resistance
- c) current
- d) power

Correct answer: a) capacitance

87.Faraday's ice pail experiment shows _____.

- a) charge resides on surface

b) inside

c) both

d) none

Correct answer: a) charge resides on surface

88. Coulomb's law: $F \propto$ _____.

a) $q_1 q_2 / r^2$

b) r^2 / q

c) q / r

d) constant

Correct answer: a) $q_1 q_2 / r^2$

89. $k = 9 \times 10^9$ in _____.

a) vacuum

b) air

c) water

d) all

Correct answer: a) vacuum

90. Electric field due to point charge _____.

a) kq/r^2

b) q/r

c) kr/q

d) zero

Correct answer: a) kq/r^2

91. Electric field direction from _____.

a) + to -

b) - to +

c) tangential

d) random

Correct answer: a) + to -

92. Electric dipole: two _____.

a) equal opposite charges

b) same charges

c) protons

d) electrons

Correct answer: a) equal opposite charges

93. Torque on dipole in uniform field _____.

a) $pE \sin\theta$

b) zero

c) p/E

d) constant

Correct answer: a) $pE \sin\theta$

94. Equipotential surface: work _____.

a) zero between points

b) maximum

c) minimum

d) infinite

Correct answer: a) zero between points

95. Electric field lines from _____.

a) + to -

b) never intersect

c) both

d) circle

Correct answer: c) both

96. Capacitance $C =$ _____.

- a) Q/V
- b) V/Q
- c) QV
- d) $1/V$

Correct answer: a) Q/V

97. Parallel plate capacitor: $C =$ _____.

- a) $\epsilon A/d$
- b) Ad/ϵ
- c) $\epsilon d/A$
- d) A/d

Correct answer: a) $\epsilon A/d$

98. Energy stored in capacitor: $\frac{1}{2}$ _____.

- a) QV
- b) CV^2
- c) Q^2/C
- d) all equivalent

Correct answer: d) all equivalent

99. Van de Graaff generator produces _____.

- a) high voltage
- b) high current
- c) low voltage
- d) AC

Correct answer: a) high voltage

100. Lightning is discharge due to _____.

- a) charge separation in clouds

b) friction

c) both

d) induction only

Correct answer: c) both